

✓ GEOG5415M Final Project

Housing affordability inequality in England (2008–2024)

Research focus: How does housing affordability (price-to-earnings) vary across space and deprivation groups, and what are the implications for public good?

Executive summary

This project examines housing affordability inequality across England using Local Authority District (LAD) data from 2008–2024. I combine the UK House Price Index (October prices), ASHE median annual earnings, and the English Indices of Multiple Deprivation (IMD 2019) to construct a price-to-earnings affordability indicator and compare patterns across deprivation quintiles. Results show clear temporal change and strong spatial unevenness in 2024, indicating that housing pressure is not evenly shared across places. These findings are relevant for local authorities and housing policy teams because they highlight where affordability pressures are concentrated and where monitoring and targeted interventions may be most needed.

1. Research question and public good motivation

Research questions:

1. How does the price-to-earnings ratio vary across deprivation quintiles in England over 2008–2024?
2. What does the spatial distribution of affordability pressure look like in 2024?

Why this matters (public good): Housing affordability affects access to stable accommodation and shapes broader inequalities (e.g., displacement pressure, labour-market access, and wealth accumulation). Place-based disparities are especially important for public good because many housing levers—planning, local housing strategies, monitoring, and local support—operate at or are reported using local authority geographies.

Primary audiences: local authorities (housing strategy and planning teams), policymakers monitoring inequality, and the general public/civil society organisations concerned with housing justice.

2. Scope note (England-only)

This project focuses on England because the Indices of Multiple Deprivation (IMD) 2019 is an England-specific deprivation framework. For conceptual consistency, I restrict the analytical sample and mapping to England LADs (LAD codes with prefixes E06/E07/E08/E09).

3. Data sources, licences, and ethics

3.1 Data inventory

I use the following open datasets:

- **UK House Price Index (UK HPI):** LAD-level October average price (P_{oct}) and derived annual log-change fields available in the provided panel.
- **ASHE (Annual Survey of Hours and Earnings):** LAD-level median annual gross earnings (residence-based, full-time).
- **English Indices of Multiple Deprivation (IMD) 2019:** LAD-level deprivation summary and deprivation quintile grouping.
- **ONS LAD boundaries (May 2024, BFC):** administrative boundary polygons used to create the 2024 map.

3.2 Ethics, bias, and representativeness

All datasets are aggregated to administrative areas and contain no personal identifiers, so privacy risks are low. However, several biases and limitations remain:

- **Measurement bias:** affordability depends on the choice of price measure (October price) and earnings definition (median annual gross, full-time, residence-based), which may not capture all households.
- **Aggregation bias / MAUP:** LAD-level statistics can mask within-area inequalities and neighbourhood-level affordability issues.
- **Time-invariance of deprivation:** IMD 2019 is used as a fixed proxy of deprivation for the full 2008–2024 period; relative deprivation may change over time.
- **Conceptual caution:** observed differences across deprivation groups are descriptive and do not imply causal mechanisms.

4. Methods overview

4.1 Data preparation and merging

I harmonise identifiers using the LAD `area_code`. I merge UK HPI with ASHE

earnings by (area_code, year), then merge the resulting table with IMD 2019 by area_code, and compute the affordability indicator:
$$\frac{P_{\text{oct}}}{\text{median annual earnings}}$$

4.2 Visual analytics choices

To compare inequality across deprivation groups, I summarise affordability using the **median** price-to-earnings within each IMD quintile per year (robust to outliers). For spatial communication, I map the 2024 price-to-earnings ratio at LAD level to support place-based interpretation for policy audiences.

5. Data quality checks

I check missingness in the key variables (P_oct, earnings, IMD quintiles) and validate the boundary merge for mapping. The 2024 join to the LAD boundary achieves a high match rate ($\approx 98.7\%$), indicating that only a small share of records are unmatched—likely due to boundary-code version differences or missing values.

6. Results

6.1 Affordability inequality over time (2008–2024)

Figure 1 shows that price-to-earnings ratios change substantially over time and differ across deprivation quintiles. This indicates that affordability pressure is not uniform: some deprivation groups consistently face higher house prices relative to local earnings. The time-series view also helps identify periods where affordability shifts coincide with broader macroeconomic phases (e.g., post-2008 adjustment and later tightening periods), although this analysis remains descriptive. Notably, the least deprived quintile (Q1) shows the highest price-to-earnings ratios across most years, consistent with higher house prices in more affluent areas relative to local earnings, while the most deprived quintile (Q5) tends to have lower ratios.

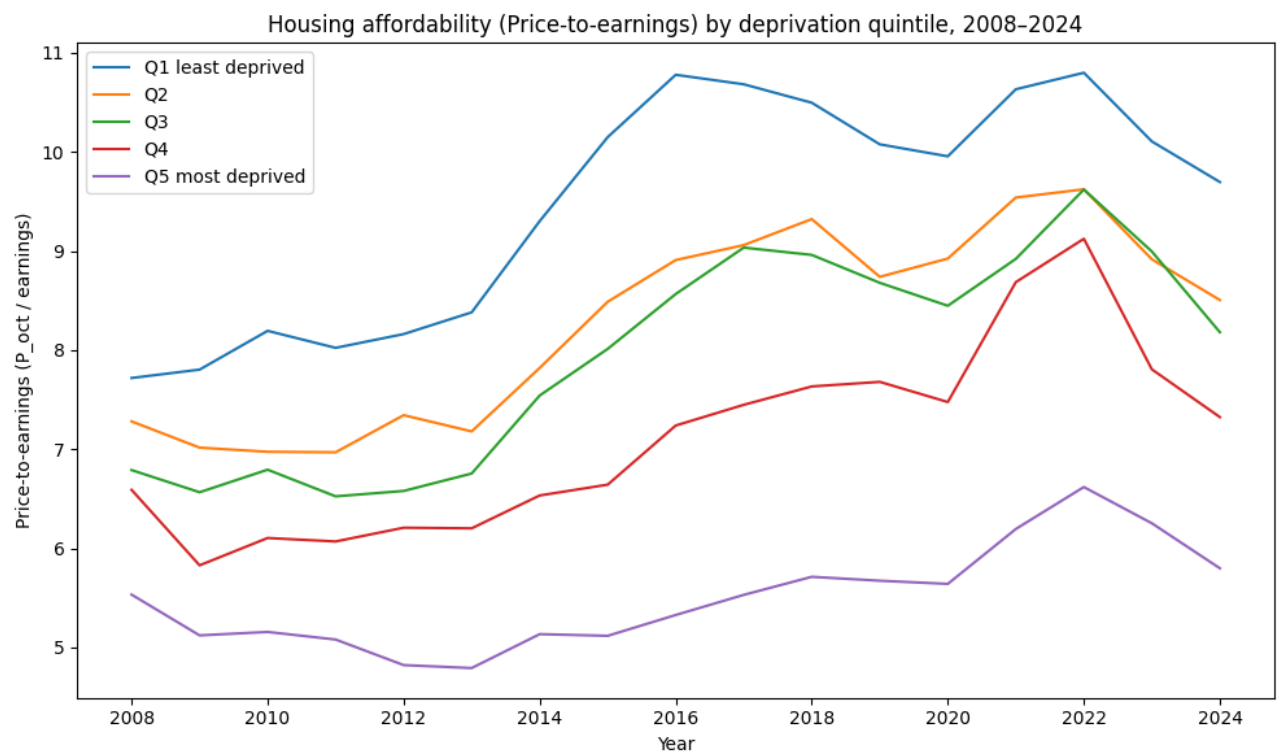


Figure 1. Housing affordability (price-to-earnings) by deprivation quintile, 2008–2024.

Audience: Local authorities and housing policy teams (and an informed public audience).

Message: Affordability (house price relative to median earnings) differs systematically across deprivation quintiles and evolves over time, indicating unequal housing pressure across places.

Design choices: I use the median price-to-earnings within each quintile for robustness to outliers; quintiles provide an intuitive inequality framing for non-technical audiences; the time-series format supports identifying structural shifts and post-shock periods.

Caveat: This is descriptive: differences may reflect multiple mechanisms (local labour markets, supply constraints, migration, composition). Aggregation to LAD can mask within-area inequalities (MAUP).

6.2 Spatial pattern of affordability in 2024

Figure 2 maps the price-to-earnings ratio across England in 2024 and highlights strong spatial unevenness. This spatial clustering is relevant for public good because it points to where affordability monitoring and targeted policy attention may be most urgent. Importantly, LAD-level mapping provides a policy-relevant lens but can hide within-area disparities.

Price-to-earnings ratio by Local Authority District (England), 2024

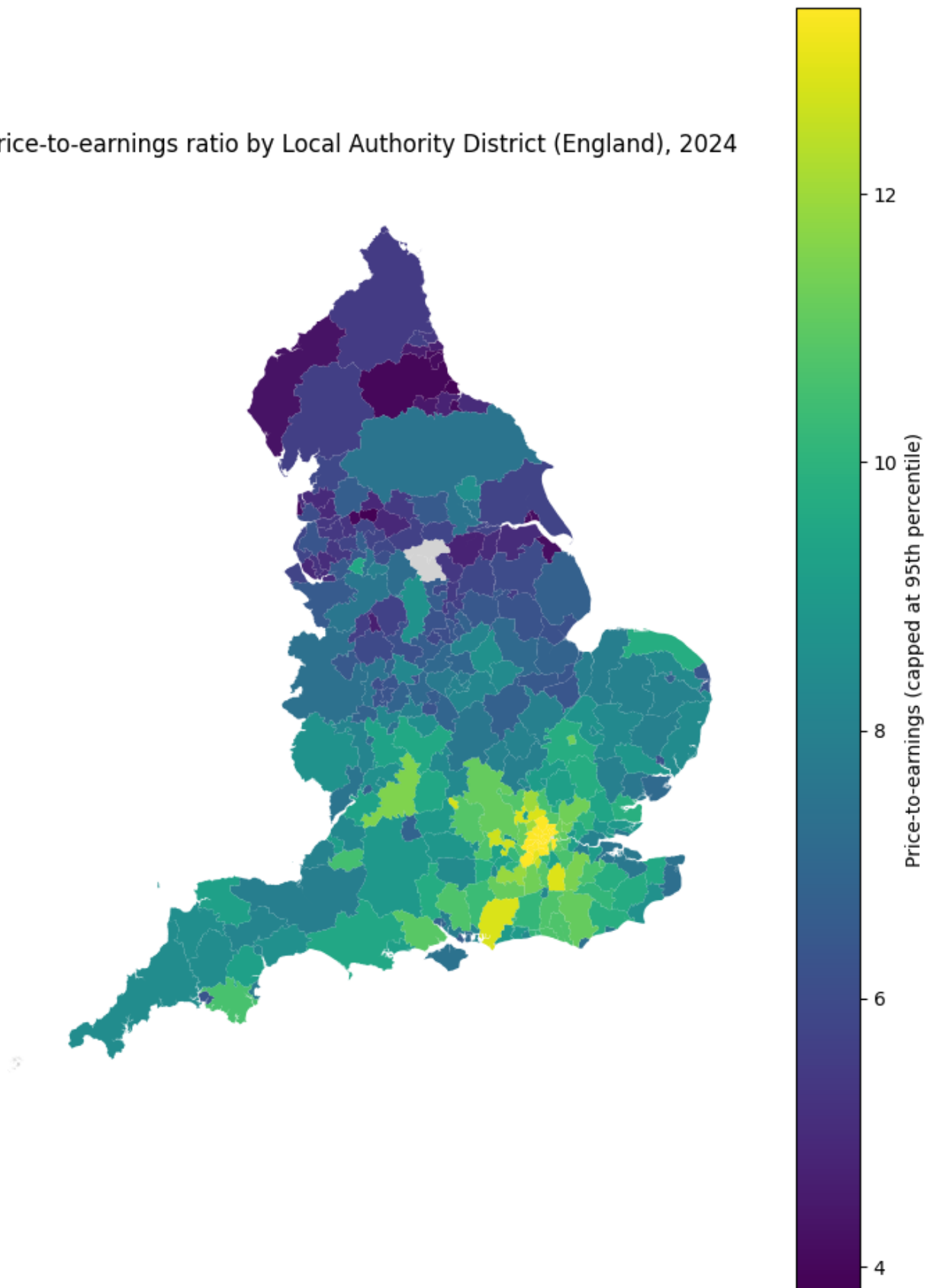


Figure 2. Price-to-earnings ratio by Local Authority District (England), 2024.

Audience: Place-based decision makers (local councils, housing strategy teams) and the general public.

Message: Housing affordability pressure is spatially uneven in 2024, highlighting areas where housing costs are high relative to local earnings. Design choices: LAD-level mapping supports policy relevance because many housing interventions

and monitoring targets align with local authority boundaries; I map price-to-earnings to communicate affordability directly; I exclude non-England LADs for conceptual consistency because IMD 2019 is an England deprivation framework.

Caveat: Boundary versions and code harmonisation lead to small unmatched cases (match rate $\approx 98.7\%$); results are sensitive to the chosen earnings measure and the October price reference. A small number of LADs have missing 2024 values (shown in grey), likely due to missing earnings after merging.

✓ 7. Discussion, limitations, and next steps

Implications for public good: The combined evidence of deprivation-group differences and 2024 spatial clustering suggests that affordability pressures are unevenly distributed and may reinforce wider spatial inequality. For local authorities, this supports prioritising affordability monitoring and aligning housing strategies with local earnings contexts rather than relying on prices alone.

Limitations: This project is descriptive and does not identify causal drivers. Affordability measurement is sensitive to indicator choices (October prices; full-time median earnings), and LAD aggregation can mask neighbourhood-level problems (MAUP). Small boundary-code mismatches also lead to a small share of unmatched records in the 2024 map ($\approx 1.3\%$).

Next steps: With more time and data, the analysis could be extended by (i) using finer spatial units (e.g., LSOA), (ii) incorporating additional constraints such as housing supply indicators or rental affordability measures, and (iii) conducting formal modelling or quasi-experimental designs to test mechanisms rather than describing association.

```
from pathlib import Path

ROOT = Path("/content/project")
DATA = ROOT / "data" / "raw"
BOUND = DATA / "boundaries"
OUT_FIG = ROOT / "outputs" / "figures"

for p in [DATA, BOUND, OUT_FIG]:
    p.mkdir(parents=True, exist_ok=True)

print("ROOT:", ROOT)
print("DATA:", DATA)
print("BOUND:", BOUND)
print("OUT_FIG:", OUT_FIG)
```

```
ROOT: /content/project
DATA: /content/project/data/raw
BOUND: /content/project/data/raw/boundaries
OUT_FIG: /content/project/outputs/figures
```

```
%%capture
!pip -q install geopandas pyogrio shapely
```

```
import pandas as pd
from pathlib import Path

DATA = Path("/content/project/data/raw")

HPI_PATH = DATA / "uk_hpi_oct_panel_y_next.csv"
EARN_PATH = DATA / "earnings_ashe_residence_fulltime_median_annual_"
IMD_PATH = DATA / "imd2019_lad_clean.csv"

hpi = pd.read_csv(HPI_PATH)
earn = pd.read_csv(EARN_PATH)
imd = pd.read_csv(IMD_PATH)

print("hpi:", hpi.shape)
print("earn:", earn.shape)
print("imd:", imd.shape)
```

```
hpi: (8850, 6)
earn: (5920, 4)
imd: (317, 8)
```

```

for _df in [hpi, earn, imd]:
    _df["area_code"] = _df["area_code"].astype(str)

hpi["year"] = pd.to_numeric(hpi["year"], errors="coerce")
earn["year"] = pd.to_numeric(earn["year"], errors="coerce")

eng_prefix = ("E06", "E07", "E08", "E09")
hpi = hpi[hpi["area_code"].str.startswith(eng_prefix)].copy()
earn = earn[earn["area_code"].str.startswith(eng_prefix)].copy()
imd = imd[imd["area_code"].str.startswith(eng_prefix)].copy()

df = hpi.merge(
    earn[["area_code", "year", "earnings_median_annual_gross"]],
    on=["area_code", "year"],
    how="left"
).merge(
    imd[["area_code", "imd_avg_score", "imd_quintile_score"]],
    on="area_code",
    how="left"
)

df["P_oct"] = pd.to_numeric(df["P_oct"], errors="coerce")
df["earnings_median_annual_gross"] = pd.to_numeric(df["earnings_med
df["price_to_earnings"] = df["P_oct"] / df["earnings_median_annual_

df = df[df["year"].between(2008, 2024)].copy()
df["imd_quintile_score"] = df["imd_quintile_score"].astype(str).str

keep_cols = ["area_code", "area_name", "year", "P_oct", "earnings_media
df = df[keep_cols].copy()

print("df shape (2008–2024):", df.shape)
print("Missing earnings (2008–2024):", df["earnings_median_annual_g
print(df.head(3).to_string(index=False))

```

```

df shape (2008–2024): (5015, 8)
Missing earnings (2008–2024): 0.06041874376869392
area_code area_name year P_oct earnings_median_annual_gross imc
E06000001 Hartlepool 2008 112602 23392.0
E06000001 Hartlepool 2009 103603 25095.0
E06000001 Hartlepool 2010 99029 23663.0

```

```

import matplotlib.pyplot as plt
from pathlib import Path
import pandas as pd

OUT_FIG = Path("/content/project/outputs/figures")
OUT_FIG.mkdir(parents=True, exist_ok=True)

df_plot = df.dropna(subset=["price_to_earnings", "imd_quintile_scor

```

```

df_plot["imd_quintile_score"] = df_plot["imd_quintile_score"].astype

order = ["Q1 least deprived", "Q2", "Q3", "Q4", "Q5 most deprived"]
df_plot["imd_quintile_score"] = pd.Categorical(df_plot["imd_quintil

g = (df_plot
      .groupby(["year", "imd_quintile_score"], observed=True)["price
      .median()
      .reset_index())

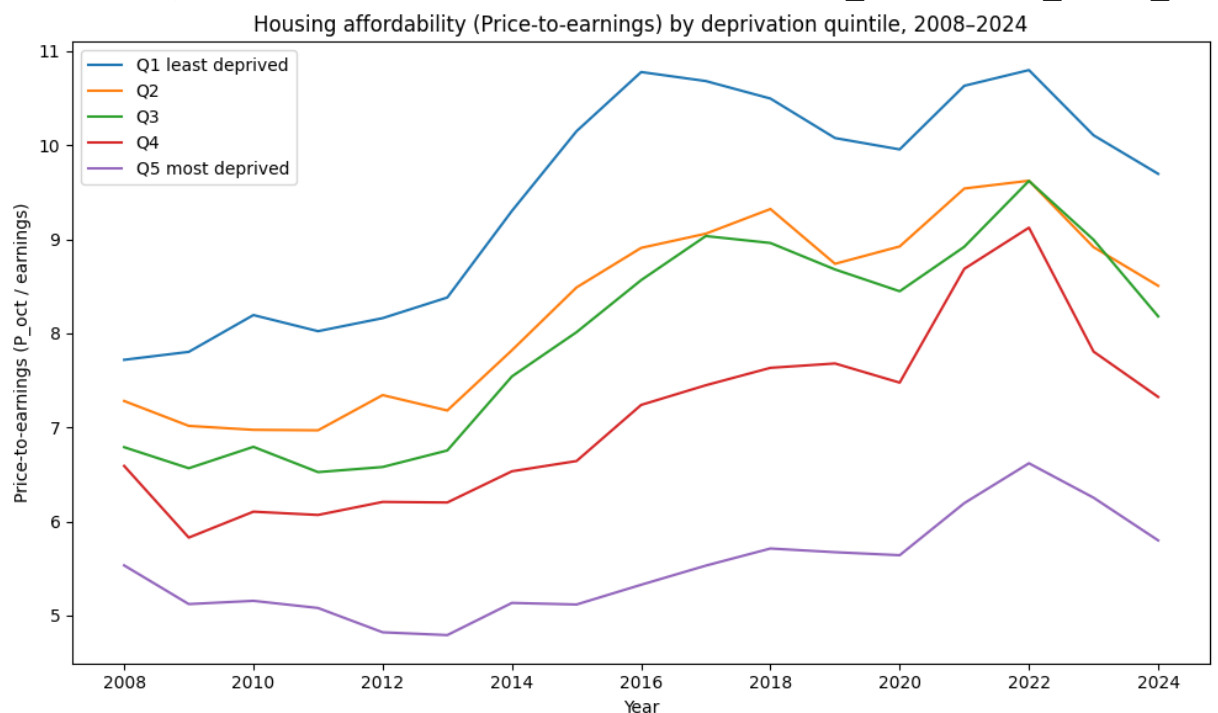
plt.figure(figsize=(10, 6))
for q in order:
    sub = g[g["imd_quintile_score"] == q]
    if sub.empty:
        print(f"Warning: no data for {q}")
        continue
    plt.plot(sub["year"], sub["price_to_earnings"], label=q)

plt.title("Housing affordability (Price-to-earnings) by deprivation
plt.xlabel("Year")
plt.ylabel("Price-to-earnings (P_oct / earnings)")
plt.legend()
plt.tight_layout()

fig1 = OUT_FIG / "fig_nonspatial_trend_imd_quintiles_2008_2024.png"
plt.savefig(fig1, dpi=300)
fig1

```

PosixPath('/content/project/outputs/figures/fig_nonspatial_trend_imd



```
import geopandas as gpd
from pathlib import Path
```

```
BOUND = Path("/content/project/data/raw/boundaries")
BOUNDARY_PATH = BOUND / "Local_Authority_Districts_May_2024_Boundar
```

```
lad = gpd.read_file(BOUNDARY_PATH)
print("Rows, cols:", lad.shape)
print("Columns:", list(lad.columns))
print(lad[["LAD24CD", "LAD24NM"]].head(5).to_string(index=False))
```

```
Rows, cols: (361, 10)
Columns: ['FID', 'LAD24CD', 'LAD24NM', 'LAD24NMW', 'BNG_E', 'BNG_N',
LAD24CD          LAD24NM
E06000001      Hartlepool
E06000002      Middlesbrough
E06000003 Redcar and Cleveland
E06000004      Stockton-on-Tees
E06000005      Darlington
```

```

candidates = [c for c in lad.columns if ("LAD" in c.upper() and c.u
print("Possible code columns:", candidates)

CODE_COL = candidates[0]
print("Using CODE_COL:", CODE_COL)

lad2 = lad.rename(columns={CODE_COL: "area_code"}).copy()
lad2 = lad2[lad2["area_code"].astype(str).str.startswith(("E06", "E0

print("England LAD rows:", lad2.shape)
lad2[["area_code"]].head()

```

```

Possible code columns: ['LAD24CD']
Using CODE_COL: LAD24CD
England LAD rows: (296, 10)

```

	area_code
0	E06000001
1	E06000002
2	E06000003
3	E06000004
4	E06000005

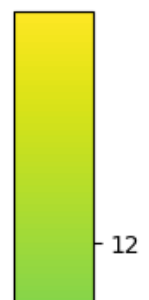
```

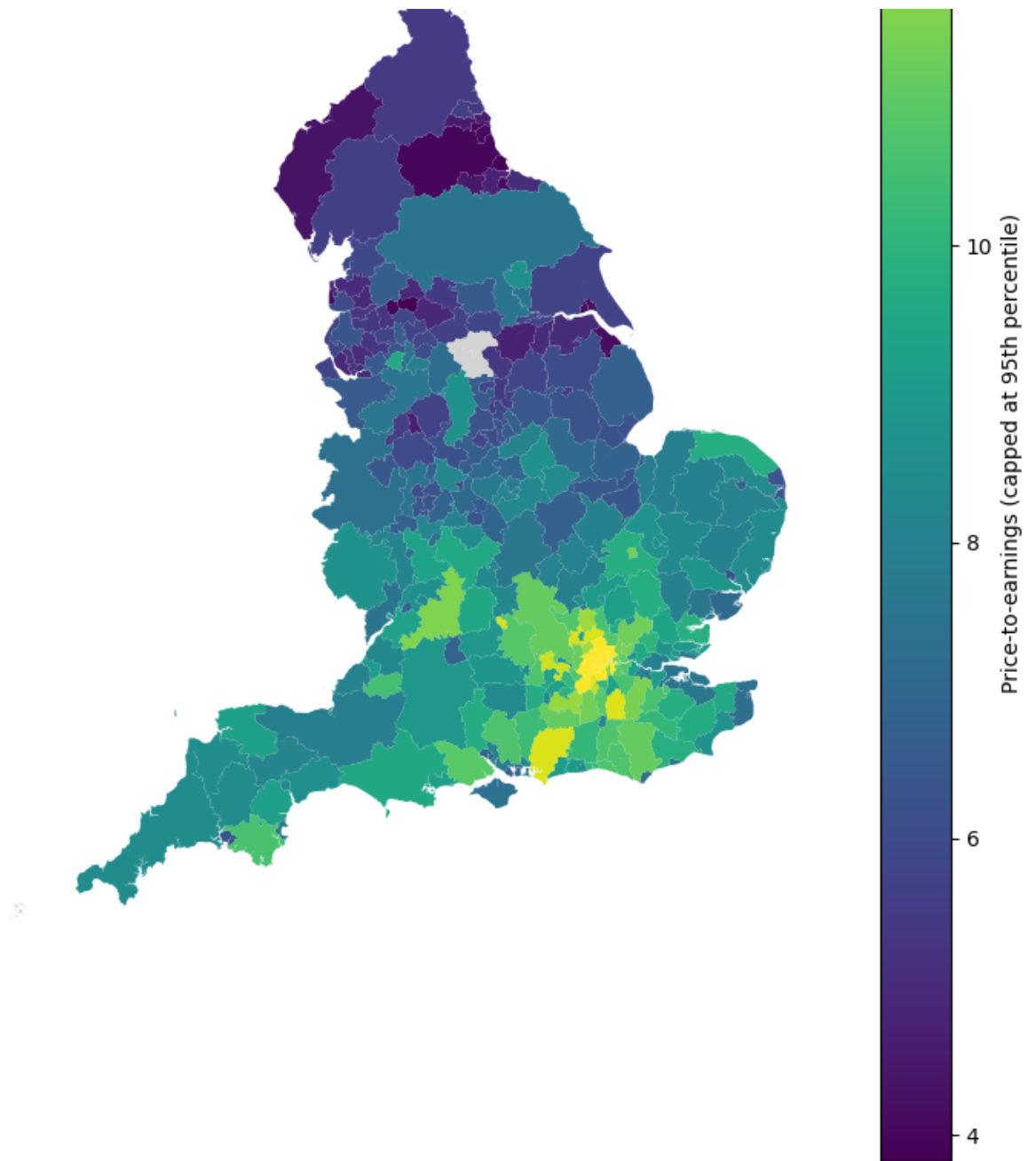
vmax = gdf2024["price_to_earnings"].quantile(0.95)

ax = gdf2024.plot(
    column="price_to_earnings",
    legend=True,
    figsize=(9,12),
    vmin=gdf2024["price_to_earnings"].min(),
    vmax=vmax,
    missing_kwds={"color": "lightgrey", "label": "Missing"}
)
ax.set_axis_off()
ax.set_title("Price-to-earnings ratio by Local Authority District (
cax = ax.get_figure().axes[-1]
cax.set_ylabel("Price-to-earnings (capped at 95th percentile)")
plt.savefig(OUT_FIG / "fig_spatial_price_to_earnings_2024.png", dpi

```

Price-to-earnings ratio by Local Authority District (England), 2024





```
missing_codes = sorted(set(lad2["area_code"]) - set(d2024["area_cod  
missing_codes[:20], len(missing_codes)
```

```
(['E06000053', 'E08000016', 'E08000019', 'E09000001'], 4)
```

```

qa = {
    "Rows (df)": int(len(df)),
    "Unique LADs (df)": int(df["area_code"].nunique()),
    "Years covered (df)": f"{int(df['year'].min())}-{int(df['year'].max())}",
    "Missing earnings rate (df)": float(df["earnings_median_annual"].isna().mean()),
    "Missing IMD quintile rate (df)": float(df["imd_quintile_score"].isna().mean()),
    "Missing P_oct rate (df)": float(df["P_oct"].isna().mean()),
}

qa["2024 match rate to LAD boundary"] = 0.9864864864864865
qa

```

```

{'Rows (df)': 5015,
 'Unique LADs (df)': 295,
 'Years covered (df)': '2008–2024',
 'Missing earnings rate (df)': 0.06041874376869392,
 'Missing IMD quintile rate (df)': 0.0,
 'Missing P_oct rate (df)': 0.0,
 '2024 match rate to LAD boundary': 0.9864864864864865}

```

References

Department for Levelling Up, Housing and Communities (2019) *English indices of deprivation 2019*. Available at: <https://www.gov.uk/government/statistics/english-indices-of-deprivation-2019> (Accessed: 28 January 2026).

HM Land Registry (2025) *UK House Price Index data downloads*. Available at: <https://www.gov.uk/government/statistical-data-sets/uk-house-price-index-data-downloads> (Accessed: 28 January 2026).

Office for National Statistics (2024) *Annual Survey of Hours and Earnings (ASHE)*. Available at: <https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/earningsandworkinghours> (Accessed: 28 January 2026).

Office for National Statistics (2024) *Local Authority Districts (May 2024) Boundaries UK BFC*. Available at: <https://geoportal.statistics.gov.uk/> (Accessed: 28 January 2026).

